Theory of Languages and Automata

Non-regular Languages

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March 2022

Countable and Uncountable Sets

Definition

A set A is said to be countable if there is an injective function $f : A \to \mathbb{N}$.

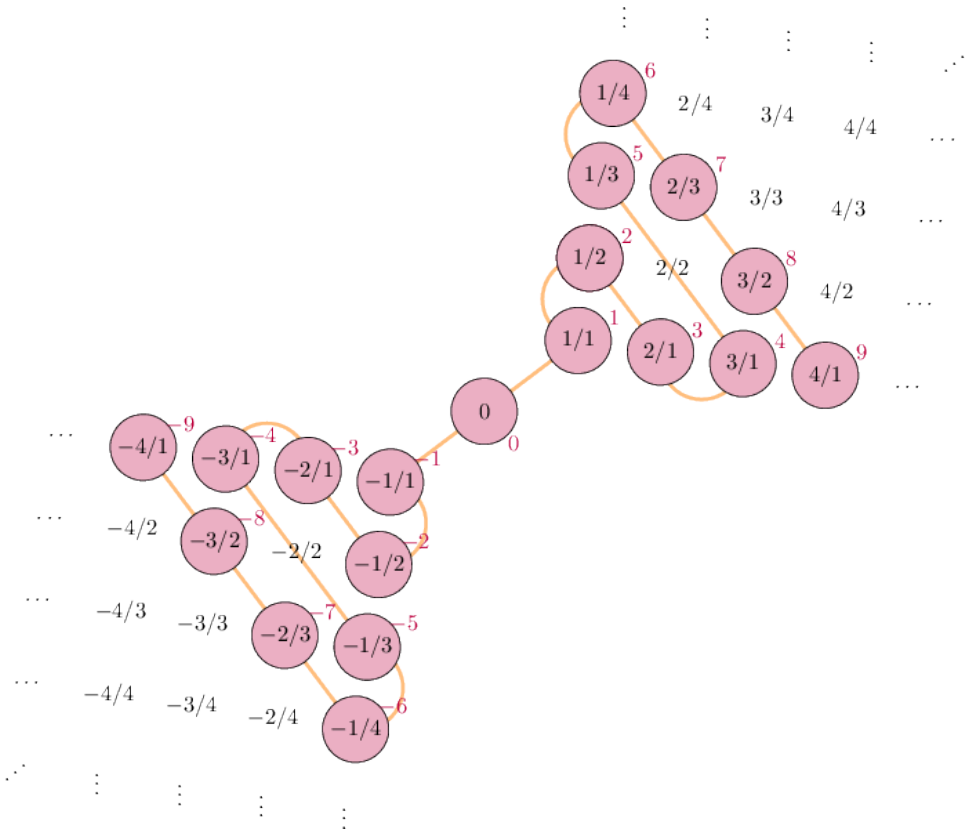
Examples:

- Any finite set
- Any subset of a countable set

Countable and Uncountable Sets

Examples:

- $\mathbb{Q}$ is countable.



- A, B are countable $\implies$ $A \times B$ is countable

Countable and Uncountable Sets

Theorem

Union of countably many countable sets is countable.

Proof.

Suppose that for every $i \in \mathbb{N}$, A_i is countable i.e. for each i, there exists an injective function

$$f_i : A_i \to \mathbb{N}.$$

Define an injective function $f : \bigcup_{i=1}^{\infty} A_i \to \mathbb{N} \times \mathbb{N}$ s.t.

$$\forall a_{i,j} \in A_i \qquad f(a_{i,j}) := (f_i(a_{i,j}), i)$$

$\square$

Countable and Uncountable Sets

How many DFAs are there over a fixed input alphabet?

Using similar techniques we used for previous statements, one can prove that:

Theorem

The set of all DFAs over a fixed alphabet is countable.

Countable and Uncountable Sets

Are there any uncountable sets?

- Contor's celebrated proof showed that uncountable sets exist.
- He used the *diagonalization* method to show that $\mathbb{R}$ is uncountable.
- Diagonalization is at the heart of several important proofs in Theory of Computation.

$$
\begin{array}{ll}
s_1 & = 0\,0\,0\,0\,0\,0\,0\,0\,0\,0\,0\ldots \\
s_2 & = 1\,1\,1\,1\,1\,1\,1\,1\,1\,1\,1\ldots \\
s_3 & = 0\,1\,0\,1\,0\,1\,0\,1\,0\,1\,0\ldots \\
s_4 & = 1\,0\,1\,0\,1\,0\,1\,0\,1\,0\,1\ldots \\
s_5 & = 1\,1\,0\,1\,0\,1\,1\,0\,1\,0\,1\ldots \\
s_6 & = 0\,0\,1\,1\,0\,1\,1\,0\,1\,1\,0\ldots \\
s_7 & = 1\,0\,0\,0\,1\,0\,0\,0\,1\,0\,0\ldots \\
s_8 & = 0\,0\,1\,1\,0\,0\,1\,1\,0\,0\,1\ldots \\
s_9 & = 1\,1\,0\,0\,1\,1\,0\,0\,1\,1\,0\ldots \\
s_{10} & = 1\,1\,0\,1\,1\,1\,0\,0\,1\,0\,1\ldots \\
s_{11} & = 1\,1\,0\,1\,0\,1\,0\,0\,1\,0\,0\ldots \\
\vdots & \quad \vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\ddots
\end{array}
$$

$$
s = 1\,0\,1\,1\,1\,0\,1\,0\,0\,1\,1\ldots
$$

Countable and Uncountable Sets

In a similar way (using diagonalization) one can prove:

Theorem

The set of all languages over a fixed finite alphabet is uncountable.

Now, we are able to give a non-constructive proof for the existence of non-regular languages.

Countable and Uncountable Sets

Theorem

Over a fixed alphabet, there are infinitely many non-regular languages.

Proof.

- There are uncountably many languages over a fixed alphabet.
- There are a countable number of DFAs over a fixed alphabet.
- There is a many-to-one correspondence from DFAs to the set of all languages.

□

Constructive proofs for existence of non-regular languages

Two ways to prove that a given language is not regular:

- Prove that if it is accepted by a DFA, then the number of states must be larger than any given number
- Pumping lemma

A Lower-bound for the Number of States

Definition

Let L be a language over an alphabet Σ. Two strings x and y are said to be *distinguishable with respect to* L if there is a string z such that $xz \in L$ and $yz \notin L$, or vice versa.

Definition

Let L be a language. A set of strings $\{x_1, \ldots, x_k\}$ is distinguishable with respect to L, if for every two distinct strings x_i and x_j, x_i is distinguishable from x_j.

Example

For $L = \{a^n b^n \ : \ n \in \mathbb{N}\}$, two strings 0 and 00 are distinguishable.

A Lower-bound for the Number of States

Theorem

Let L be a language, and suppose there exist a set of k distinguishable strings with respect to L. Then every DFA accepting L has at least k states.

Proof.

Proof by contradiction:

- By pigeonhole principle there exist $x_i \neq x_j$ for which:

$$\hat{\delta}(q_0, x_i) = \hat{\delta}(q_0, x_j)$$

- Therefore, x_i and x_j are not distinguishable.

□

A Lower-bound for the Number of States

Example

By the above theorem, for the following languages, if an accepting DFA exists, it should have infinitely many states.

- $\texttt{Pal} = \{ww^r \ : \ w \in \Sigma^*\}$
  Since every two distinct strings $x, y \in \Sigma^*$ are distinguishable.
- $\mathcal{L} = \{a^n b^n \ : \ n \in \mathbb{N}\}$
  The set $\{a^i \ : \ i \in \mathbb{N}\}$ is distinguishable with respect to $\mathcal{L}$.
- $\mathcal{L}' = \{a^{n^3} \ : \ n \in \mathbb{N}\}$
  Same as the previous example.

Hence these languages are not regular.

Pumping Lemma

Theorem

For every regular language L, there is an integer n, such that for every string $w \in L$ of length $\geq n$, there exist strings $x, y, z \in \Sigma^*$ s.t.

- $|xy| \leq n$.
- $|y| > 0$.
- *For all* $i \geq 0$, $xy^iz \in L$.

Below is a formal expression of the pumping lemma:

$$(\forall L \subseteq \Sigma^*)$$
$$(regular(L) \implies$$
$$((\exists p \geq 1)((\forall \omega \in L)((|\omega| \geq p) \implies$$
$$((\exists x, y, z \in \Sigma^*)(\omega = xyz \wedge (|y| \geq 1 \wedge |xy| \leq p \wedge (\forall n \geq 0)(xy^nz \in L))))))$$

Pumping Lemma

Example

$\mathcal{L} = \{a^i b^i \ : \ i \text{ is a square number}\}$ is not regular.

- Suppose it were; then there would be an associated n for the pumping lemma.
- Let $w = a^{n^2} b^{n^2}$. If we write $w = xyz$, with $|xy| \leq n$ and $y \neq \epsilon$, then $y = a^k$ for some $k \geq 1$.
- Then $xy^iz = a^{n^2+ik} b^{n^2} \in L$, but for every $i \geq 1$, xy^iz can't be written as $a^{m^2} b^{m^2}$ for any $m \in \mathbb{N}$.

Pumping Lemma

Example

$\mathcal{L}' = \{a^i b^j \ : \ i > j\}$ is not regular.

- Suppose it were. Then there would be an associated n for the pumping lemma.
- Let $w = a^{n+1} b^n$. We can write $w = xyz$, and by conditions on $|xy|$ and $|y|$ in the pumping lemma, x and y consist of a's only and $y \neq \epsilon$.
- Let $i = 0$. Since $xy^i z \in \mathcal{L}'$ for every $i \geq 0$, $xz \in \mathcal{L}'$, but this string has fewer a's than w.